

$$1) \quad q_c(t) = m(t) \cos(\omega_c t_0) - \hat{m}(t) \sin(\omega_c t_0) +$$

$$\Rightarrow q_{\Gamma} = m(t), \quad q_Q = \hat{m}(t)$$

$$\Rightarrow g(t) = \tilde{a}_e(t) = m_+(t)$$

$$m_+(t) = m(t) + j \hat{m}(t)$$

$$\rightarrow \hat{M}(f) = -j \operatorname{sgn}(f) \operatorname{tr}(f)$$

$$\rightarrow Z(t) = m_+(t) e^{j2\pi f_0 t} = (m_+ + j\hat{m}(t)) (\cos(2\pi f_0 t) - j \sin(2\pi f_0 t))$$

$$2) \quad F\{q_e(t)\} = \underbrace{F\{m(t) \cos(2\pi f_0 t)\}}_{I_1} - \underbrace{F\{\hat{m}(t) \sin(2\pi f_0 t)\}}_{I_2}$$

$$I_1 = F\{m(t)\} * F\{\cos(2\pi f_0 t)\} = \frac{1}{2} (M(f-f_0) + M(f+f_0))$$

$$I_2 = \frac{1}{2j} [M(f-f_0) - M(f+f_0)], \quad \hat{m}(t) = -j \operatorname{sgn}(f) m(f)$$

$$\rightarrow I_2 = -\frac{1}{2} [M(f-f_0) - \hat{M}(f+f_0)]$$

$$\rightarrow X(f) = \frac{1}{2} \left[(1 + \operatorname{sgn}(f-f_0)) M(f-f_0) + (1 - \operatorname{sgn}(f+f_0)) M(f+f_0) \right]$$

$$= u(f-f_0) M(f-f_0) + u(f_0-f) M(f+f_0), \quad m(t): \begin{array}{c} \triangle \\ -1 \quad 1 \end{array}$$

$$\rightarrow x(t): \begin{array}{c} \triangle \\ -f_0 \end{array} \quad \begin{array}{c} \triangle \\ f_0 \quad f_0+1 \end{array} \quad \rightarrow \text{bandwidth} = 1 \text{ Hz}$$

$$3) \quad q_e(t) = m(t) \cos(2\pi f_0 t) + \hat{m}(t) \sin(2\pi f_0 t)$$

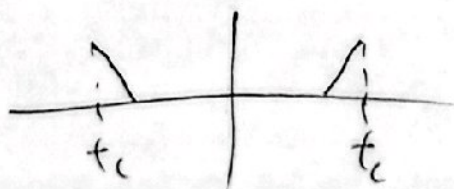
$$\rightarrow q_{eI} = m(t), \quad q_{eQ} = -\hat{m}(t)$$

$$z(t) = \tilde{q}_e(t) = q_{eI} + j q_{eQ} = m(t) - j \hat{m}(t)$$

$$g(t) = q_+^*(t) = \tilde{q}_e^* e^{j 2\pi f_0 t} = (m(t) - j \hat{m}(t)) e^{j 2\pi f_0 t}$$

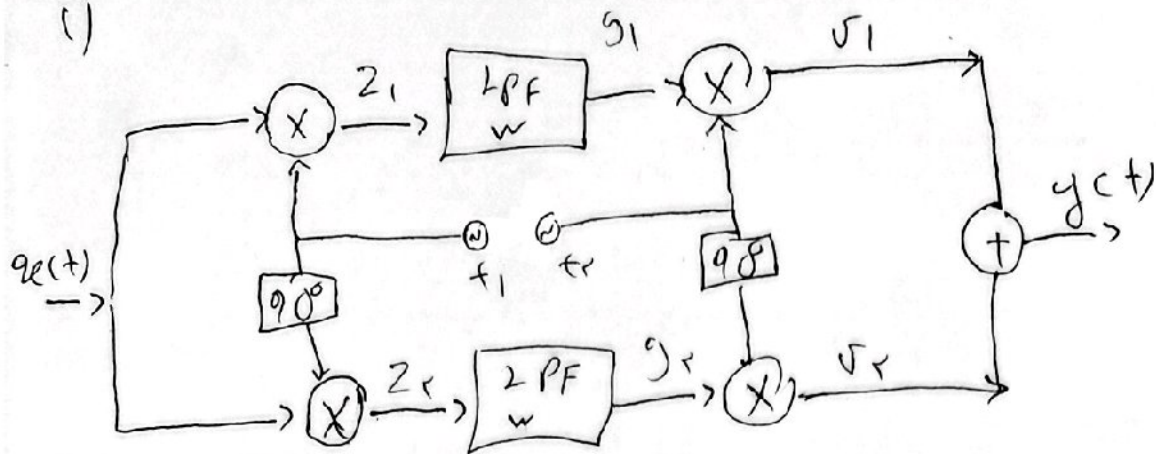
$$X(f) = \frac{1}{2} [u(f_0-f) M(f-f_0) + u(f_0+f) M(f+f_0)]$$

same pipeline as part 2



2 weaver's SSB modulator

1)



$$z_1(t) = \cos(2\pi f_m t) \cos(2\pi f_c t) = \frac{1}{2} [\cos 2\pi(f_c - f_m)t + \cos 2\pi(f_c + f_m)t]$$

$$z_r(t) = \cos(2\pi f_m t) \sin(2\pi f_c t) = \frac{1}{2} [\sin 2\pi(f_c - f_m)t + \sin 2\pi(f_c + f_m)t]$$

→ if we put ω as: $|f_c - f_m| < \omega < f_c + f_m$

$$\Rightarrow g_1(t) = \cos 2\pi(f_c - f_m)t, \quad g_r(t) = \sin 2\pi(f_c - f_m)t$$

$$\Rightarrow v_1(t) = \cos 2\pi(f_c - f_m)t + \cos 2\pi f_c t, \quad v_r(t) = -\sin 2\pi(f_c - f_m)t + \sin 2\pi f_c t$$

$$\cos A \cos B - \sin A \sin B = \cos A + B$$

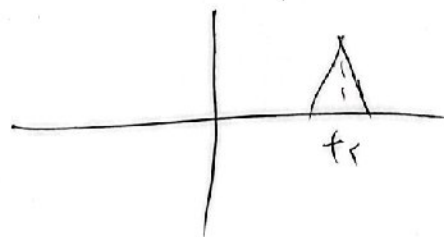
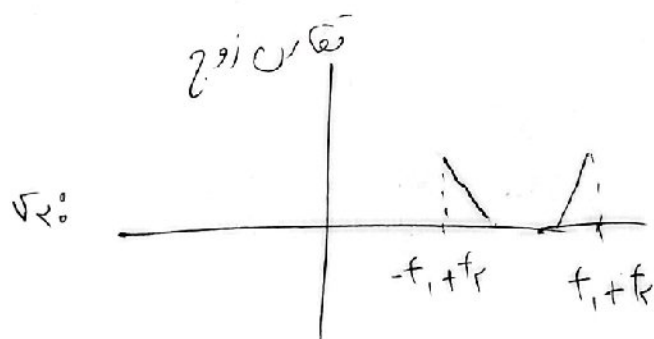
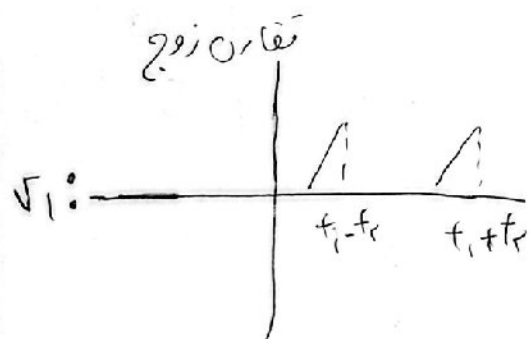
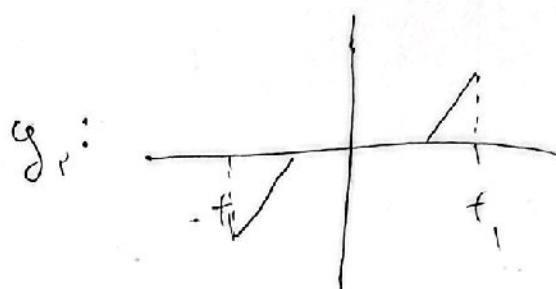
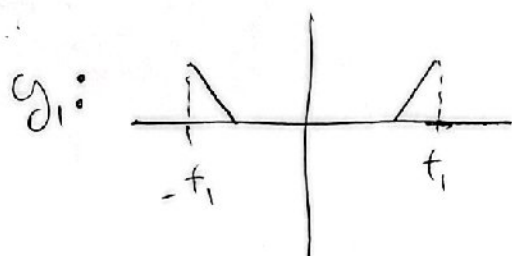
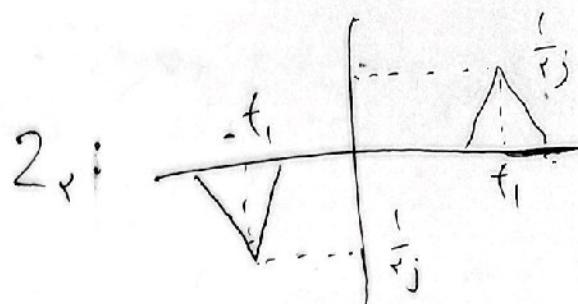
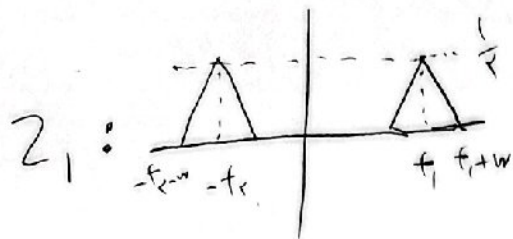
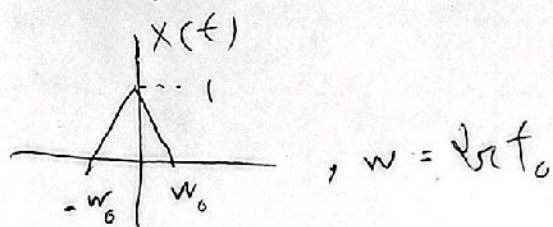
$$\Rightarrow y = \cos 2\pi(f_c + f_r - f_m)t +$$

→ if $f_c + f_r > f_m \rightarrow$ USB

2. with $f_c + f_r > f_m$ USB, we could choose

$f_c + f_r < f_m$ so we had LSB

for a general input x , and $F\{x\} = X(f)$



4 input: $x(t)$, only nonzero in $-W < t < W$

first mixer: two copies of $x(t)$ in $(f_1 + f_m)$ - $(f_1 + f_m)$

LPF: passes components at $(f_1 - f_m)$ - $(f_1 - f_m)$

Second mixer:

two copies in $f_1 + f_2 - f_m$, $-(f_1 + f_2 - f_m)$

y : single frequency at f_r

$$5 \quad y(t) = Y_{\text{upper}} \cos \omega_r t - Y_{\text{lower}} \sin(\omega_r t)$$

$$\rightarrow y(t) = \cos \omega_r (t_r + t_1 - t_m) + \sin \omega_r (t_r + t_1 - t_m) +$$

$$= \cos \omega_r (t_1 + t_r - t_m) + -90^\circ$$

3 Modulated waves

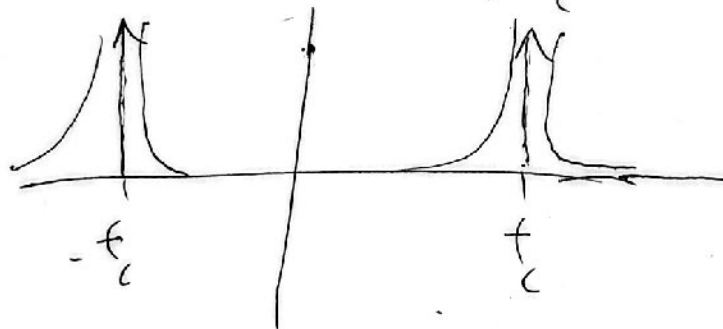
$$1) \quad m(t) = \frac{1}{1+t^2}$$

$$e^{-j\omega t} \xrightarrow[\omega \neq 1]{f.T} \frac{1}{\omega+1} \xrightarrow{\text{duality}} m(t) = \pi e^{-\pi |t|}$$

$$x_c(t) = A_c (1 + u m(t)) \cos \pi f_c t, \quad A_c = 1, u = 0/\omega$$

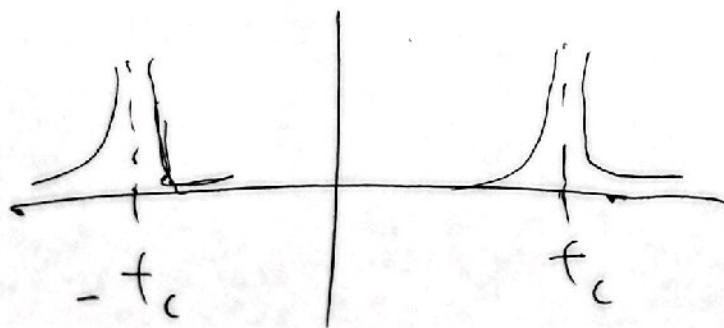
$$\rightarrow x_c(t) = \left(1 + \frac{0/\omega}{1+t^2}\right) \cos(\pi f_c t)$$

$$\rightarrow X_c(f) = \frac{1}{f} (\delta(f-f_c) + \delta(f+f_c)) + \frac{1}{f} (m(f-f_c) + m(f+f_c))$$



$$2) \quad x_c = A_c m(t) \cos 2\pi f_c t, \quad A_c = 1$$

$$\rightarrow X_c(f) = \frac{1}{f} [m(f+f_c) + m(f-f_c)]$$



4 Envelope Analysis of a VSB signal

$$g(t) = \frac{\alpha}{r} \cos(r\pi(t_c + t_m)) + \frac{1-\alpha}{r} \cos(r\pi(t_c - t_m)) + \cos r\pi t_c +$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$\rightarrow g(t) = (1 + \frac{1}{r} \cos r\pi t_m) / \cos r\pi t_c + (\frac{1-\alpha}{r} \sin r\pi t_m) / \sin(r\pi t_c)$$

$$\tilde{g}(t) = g_I(t) + j g_Q(t) = g_I \cos r\pi t_c + - g_Q \sin r\pi t_c$$

$$\rightarrow g_I(t) = 1 + \frac{1}{r} \cos r\pi t_m, \quad g_Q(t) = \frac{1-\alpha}{r} \sin r\pi t_m$$

$$\rightarrow \text{envelope} = \sqrt{g_I^2 + g_Q^2} = \sqrt{\left(1 + \frac{1}{r} \cos r\pi t_m\right)^2 + \left(\frac{1-\alpha}{r}\right)^2 \sin^2 r\pi t_m}$$

$$= \sqrt{\left(1 + \frac{1}{r} \cos r\pi t_m\right)^2 \left[1 + \frac{(1-\alpha)^2 \sin^2 r\pi t_m}{(r + \cos r\pi t_m)^2}\right]}$$

$$= \left[1 + \frac{1}{r} \cos r\pi t_m\right] \sqrt{1 + \frac{(1-\alpha)^2 \sin^2 r\pi t_m}{(r + \cos r\pi t_m)^2}}$$

$$d(t)$$

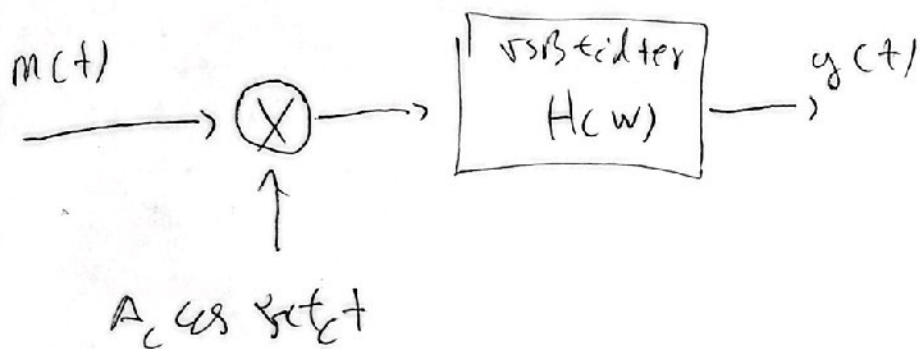
$$0 \leq \left(\frac{(1-\alpha) \sin \gamma t_m t}{r} \right) \leq \left(\frac{1-\alpha}{r} \right)^r$$

$$\rightarrow 1 \leq d(t) \leq \sqrt{1 + \left(\frac{1-\alpha}{r} \right)^r}, \quad 0 \leq \alpha < 1$$

$$\Rightarrow d_{\max} \text{ occurs when } \alpha = 0, d_{\max} = \sqrt{\frac{a}{r}}$$

5 VSB Modulation Design

1)



$$2 \quad H(\omega_c + \Omega) + H(\omega_c - \Omega) = \text{constant} \quad \text{for } |\Omega| \leq W$$

$\rightarrow H_A \rightarrow$ upper side band

$\rightarrow H_B \rightarrow$ lower side band

$\rightarrow H_C \rightarrow$ is not a valid VSB filter

6 Interpreting Modulated signals

1)

AM: $x_c(t) = (1 + a m(t)) \cos(2\pi f_c t)$, $a = \frac{1}{r}$

$\rightarrow E_{\text{envelope}} = A_c |1 + \frac{1}{r} m(t)|$, $0.6 < 1 + \frac{1}{r} m(t) < 1.6$

So the envelope is a shifted smooth version of m

$\rightarrow B: AM$

DSB: $x_c(t) = A_c \cos(2\pi f_c t + m(t))$

$E_{\text{envelope}} = A_c |m(t)| \rightarrow$ No offset or shifting

$\rightarrow$ envelope can get to zero when $m(t)$ does

$\rightarrow C: DSB$

A: sth else D: sth else, not even using $m(t)$

2) i) conventional AM: $x_c = A_c (1 + a m(t)) \cos(2\pi f_c t)$

the message is directly present at the envelope
detection only requires a simple diode envelope
detector, cheapest Analog receive possible!

ii) SSB

SSB only uses one side band

→ bandwidth per voice channel: 3^{KHz} instead of 6

also spectral efficiency is highest among

analog AM methods $\frac{4000 \text{ MHz}}{3 \text{ Hz}} = 1.33 \times 10^6 \text{ channels/s}$